

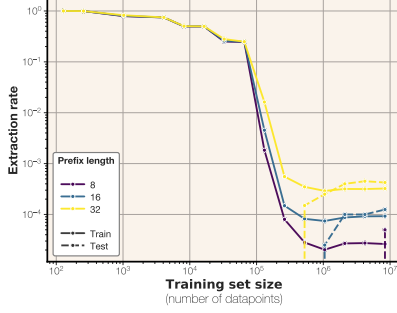


Figure 8 Extraction rates of 64-token training sequences across prefix lengths, for both train and evaluation.

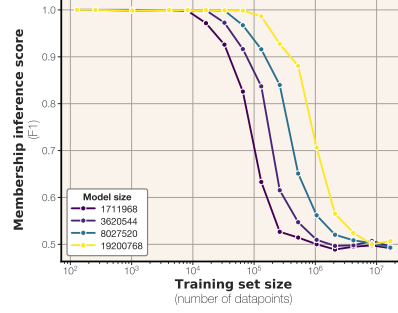


Figure 9 Membership inference F1 across dataset sizes. In this case, F1 score of 0.5 implies random guessing.

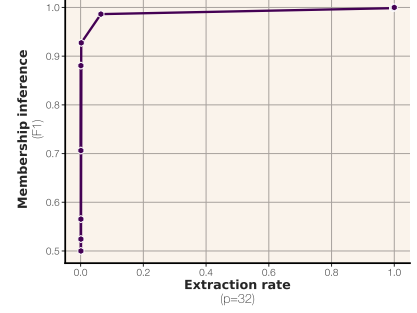


Figure 10 Membership inference vs 32-token-prefix suffix extraction rate. Membership inference is generally easier than extraction.

memorization of each model-dataset pair. In addition to memorization, we measure membership inference performance according to a standard loss-based membership inference procedure; we also compute exact extraction rates by greedily decoding prefixes of different lengths.

Results. We first observe that the sample-level unintended memorization increases with model parameters and decreases with training set size (Figure 4). When we measure unintended memorization with respect to an oracle reference model (Figure 2), memorization steadily increases as our smaller model is able to learn more about the small training set than the oracle, and then decreases as our model starts to generalize and perform on average worse than the (higher-capacity) oracle.

Dataset-to-capacity ratio predicts double descent. We observe from the train and test loss that for larger datasets the model only begins to generalize (i.e. evaluation loss decreases) once its capacity is reached, which takes approximately 10^5 samples, depending on parameter count. As in [Nakkiran et al. \(2019\)](#) we plot the ratio between the dataset size and model capacity (Figure 3). Unlike prior work, in our experiments we can compute the exact dataset size (based on the compression rates of the reference model) and exact model capacity (based on our estimate of α).

We clearly observe double descent evaluation performance decreases as the training set size nears model capacity, and then rapidly drops as the dataset capacity exceeds the capacity of the model. Our observations offer an intuitive explanation for double descent ([Belkin et al., 2019](#); [Nakkiran et al., 2019](#)): **double descent begins exactly when the data capacity exceeds the model capacity.** One theory is that once the model can no longer memorize datapoints individually, it is forced to share information between datapoints to save capacity, which leads to generalization.

Generalization explains nonzero extraction rates. We measure extraction rates on the full training set and 10,000 non-overlapping test samples (Figure 17). We note that for 32-token prefixes, 100% are extractable for very small training set sizes; predictably, all extraction numbers decrease with training set size. When the dataset sizes grows sufficiently large, the extraction rate does not go fully to zero; however, it converges to nearly exactly the test extraction rate. In other words, when our (deduplicated) dataset grows sufficiently large, **all successful training data extraction is attributable to generalization.**

5 Memorization and Membership

Our training settings allow total control over the train and test data and come with perfect deduplication. This makes our setting ideal for studying the relationship between model size, dataset size, and membership inference success rate.

All of our membership inference results come from a standard loss-based membership inference ([Yeom et al., 2018](#); [Sablayrolles et al., 2019](#)). The method is very simple: we set a cutoff loss value to predict whether a

sample is or is not a member of the training dataset.

5.1 Membership in synthetic and text data

Synthetic data. For each of our models trained on synthetic data, we plot the success rate of the membership inference attack across dataset sizes. We show results in Figure 14. Above a certain dataset size, membership inference starts to fail in the average case. This finding indicates that if the dataset size is too large compared to the model, membership inference of an average training sample may not be possible.

Text. For each of our models trained on text, we use unused non-overlapping data from FineWeb to perform a standard loss-based membership inference (Yeom et al., 2018; Sablayrolles et al., 2019) on each model and plot performance across dataset sizes (9). For a fixed model size, membership inference gets more difficult as the size of the data increases. When comparing membership inference to extraction (Figure 10), membership inference is strictly higher in every case; in some cases we can infer training dataset membership quite well (score of 0.97) with an extraction rate of 0.

5.2 Scaling laws for Membership

In this section we develop a set of predictive models for memorization. Specifically, we predict the F1 score of a loss-based membership attack given token count, number of examples, and model parameter count. We then validate our predictions on models from 500K to 1.5B parameters.

5.2.1 Functional forms

We observe that for a fixed model capacity, membership inference follows a roughly sigmoidal form with respect to dataset size. The intuitive explanation is that M.I. is easy for large models overfit to tiny datasets, so its score begin at 1; as dataset size increases, differentiating train from test data by loss becomes more and more difficult, eventually decaying toward 0.5.

We reuse the data collected in our text experiments (Section 4) to solve for constants c_1, c_2, c_3 in the following equation:

$$\text{Membership}_{F_1}(\theta, \mathcal{D}) = \frac{1}{2}(1 + c_1\sigma(c_2(\frac{\text{Capacity}(\theta)}{|\mathcal{D}|} + c_3)))$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$.

Limiting behavior. We observe that as $|\mathcal{D}| \rightarrow \infty$, performance of our membership inference attack decreases to 0.5 (essentially random performance). For a model trained on an infinite dataset, our law predicts both membership inference and extraction to be impossible.

Fitting. We use a non-linear least squares solver to find optimal values for c_1, c_2, c_3 . Solutions found are $c_1 = 1.34$, $c_2 = -0.034$, and -33.14 . We plot the scaling laws along with observed data in Figure 7. Although the sigmoidal function is slightly simplistic (the points do not perfectly fit) our fit produces estimates within 1 – 2% of observations.

5.2.2 Validation on larger models

We note that all contemporary language models trained with a tokens-per-parameter ratio of 10^2 or higher, which according to our laws would imply membership inference score of 0.5 – that is, within our formulation, statistically significant loss-based membership inference is not possible.

To validate our predictions, we train models with expected membership F_1 scores of 0.55, 0.75, and 0.95. For model sizes we select GPT-2 small (125M params) and GPT-2 XL (1.5B params). Using our scaling law, we solve for the dataset size required to get the desired membership inference score for the given model size (see Table 2 for more information). We train models on the estimated dataset size and measure F1 score (shown as circles in Figure 7).

	d_{emb}	n_{layer}	$ \theta $	$ D $	Predicted F1	Observed F1
GPT2-XL	1600	48	1,556,075,200	170,654,583	0.55	54.61 ± 1.3
				76,795,021	0.75	71.08 ± 0.4
				18,851,574	0.95	95.85 ± 0.8
GPT2-Medium	768	12	123,702,528	13,566,442	0.55	53.44 ± 1.1
				6,104,935	0.75	65.69 ± 0.6
				1,498,634	0.95	97.98 ± 0.3

Table 2 Dataset sizes that our scaling law predicts will produce a given membership inference F1, along with empirical values.

Our predictions are generally within 1.5 points of the true F1 score; the score is most inaccurate for estimated F1 of 0.75, which is the point where the sigmoid is steepest. In general, the accuracy of our results indicates that our empirical model of membership inference is relatively accurate and provides evidence for why membership inference attacks fail on models trained on extremely large datasets (Das et al., 2024; Duan et al., 2024; Maini et al., 2024).

6 Related Work

Language models and compression. Shannon’s source coding theorem (Shannon, 1948) first formalized the duality between prediction and compression. The connection between language modeling and compression was studied as far back as Shannon (1950), which observed that more accurate models of English can compress text in fewer bits. Other works note the connection between Kolmogorov complexity (Kolmogorov, 1965) and Shannon information in detail (Grunwald & Vitanyi, 2004). Delétang et al. (2024) investigate using modern transformer-based language models as compressors. We use compression as a tool to measure memorization in models.

Language model capacity. Early research on single-layer perceptrons found that single-layer networks can store up to 2 bits-per-parameter (Cover, 1965; Gardner, 1988; Baldi & Hornik, 1989). (Arpit et al., 2017) formalize the idea of *effective capacity* of a model and its training procedure; they also observe that both representation capacity and training time have a strong impact on empirical model capacity. Several other works measure language model capacity in the number of facts or random labels that can be memorized by a network such as an RNN (Collins et al., 2017; Boo et al., 2019) or transformer (Roberts et al., 2020; Heinzerling & Inui, 2021; Allen-Zhu & Li, 2024), sometimes under quantization. A few research efforts (Yun et al., 2019; Curth et al., 2023; Mahdavi et al., 2024; Kajitsuka & Sato, 2024) have developed theoretical estimates for the capacity of different model architectures, although none have yet scaled to multi-layer modern transformers. (Shwartz-Ziv et al., 2024) also analyze the ‘capacity’ of neural networks on datasets of varying size. We are the first to measure a clear upper-bound in model capacity using per-sample entropy measurements.

Information regularization in learning theory. Several works have explored the role of mutual information between the input and output of a learning algorithm (Bassily et al., 2018; Haghifam et al., 2020; Steinke & Zakynthinou, 2020). This concept closely relates to the notion of memorization based on Shannon information, discussed in Section 2.1. Some of our findings also relate to the discovery of *double descent* in machine learning (Belkin et al., 2019; Nakkiran et al., 2019) and language modeling (Xia et al., 2023), as well as general discussions of memorization and generalization in deep learning (Zhang et al., 2017; Tănzer et al., 2022).

Alternative definitions of memorization. Unintended memorization is deeply related to the many other definitions of memorization proposed in the literature. We provide a detailed comparison in the following subsections.

Prior definitions of memorization. Carlini et al. (2019) defined a string m as memorized by a language model θ if the second half of m can be generated greedily when prompting the model with the first half. Following this, Nasr et al. (2023) introduced *extractable memorization*, where model θ is said to memorize m if an adversarial prompt p can be found that generates m . Mireshghallah et al. (2022) and Schwarzschild et al. (2024) refined this definition by restricting p to a certain number of tokens, preventing it from containing the entire m . However, even this definition has limitations: for example, generating the sequence “cat cat cat ... cat” with the prompt “repeat cat 1000 times” does not necessarily indicate memorization. Carlini et al. (2019) use perplexity or likelihood, one measure of the compressibility of a sequence, in an effort to distinguish highly memorized sequences from merely easy-to-compress ones. One additional definition of note is *counterfactual memorization* (Zhang et al., 2023), which measures the impact of a single datapoint on training; this can be seen as an instantiation of our definition where a different model of the same family is used as a reference model. Overall, all these works regarded memorization in terms that can be seen as forms of compression, although did not explicitly define it as such.

Finally, a concurrent work (Cohen et al., 2024) proposes a theoretical definition for memorization also relying on Kolmogorov.

7 Conclusion

We propose a new definition of memorization that allows us to measure the exact number of bits a model knows about a dataset. We use our definition to measure the capacity of modern transformer language models and analyze how measurements such as extraction and F1 score scale with model and dataset size. We also propose a scaling law for membership inference and validate it on larger models. Our results help further practitioner understanding of how language models memorize and what they might (or might not) be memorizing across model and dataset scales.

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